

Mathematics Class 11

FBISE Complete Solution of 1st Annual Exam, 2026

Time allowed: 2:35 Hours

Total Marks Sections B and C: 80

SECTION – B (Marks 48)

Q.2 Solve the following parts.

(12 x 4 = 48)

(Note: Use of the graph paper is not necessary. Candidates can make their own grid on answer book)					
(i)	Find the sum of the geometric series $\frac{2}{3}, -1, \frac{3}{2}, \dots$ up to 5 terms.	04	OR	Sketch the graph of $y = 2\cos x$ for $-\pi \leq x \leq \pi$.	04
(ii)	Find the last two digits of $(12)^{11}$.	04	OR	Find the rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$	04
(iii)	Given $z_1 = 2(\cos 30^\circ + i \sin 30^\circ)$ and $z_2 = 3(\cos 60^\circ + i \sin 60^\circ)$, find $z_1 z_2$ in polar form.	04	OR	Find the value of 'n' if $P_1 : P_2 : P_3 = 9 : 1$	04
(iv)	Find the remainder when 9^{20} is divided by 31 using Binomial expansion.	04	OR	Find all roots of: $x^3 - 3x^2 - 4x + 12$ by Factor theorem.	04
(v)	If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ then find A_2, A_3, A_1 and A_0 .	04	OR	Prove by mathematical induction that: $2^n \geq n + 1$ for all $n \geq 1$	04
(vi)	Solve the following equations: $(2 + i)\omega = 1$ $(3 - i)z + 5\omega = 2 - i$	04	OR	A force $P = 3i - 2j + k$ acts at point $(1, 2, 3)$. Find the moment about the origin.	04
(vii)	Verify the identity: $\cos 3\alpha = 4\cos^3 \alpha - 3\cos \alpha$	04	OR	A class has 5 boys and 4 girls. In how many ways can a group of 3 students be formed?	04
(viii)	Find the volume of the parallelepiped: $\vec{a} = i + j + 2k, \vec{b} = 2i + j - k, \vec{c} = i + k$	04	OR	Find the maximum and minimum values of: $y = \frac{1}{4 + 5\cos 3x}$	2+2
(ix)	Determine whether $(x+2)$ is a factor of: $f(x) = x^3 + 2x^2 - 4x - 8$	04	OR	Find the sum to n terms of the sequence: 3, 9, 27, 81, ...	04
(x)	Determine the properties of the graph of: $y = \cos x$ i.e. Domain, Range, Period and Amplitude.	04	OR	Find the value of λ so that vectors $\vec{v} = i - j + k, \vec{u} = 2i + j + k$ and $\vec{w} = -i + \lambda j + 2k$ are coplanar.	04
(xi)	Find the Sum of first 20 terms of A.P with $a = 1, d = 3$.	04	OR	Verify: $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$	04
(xii)	Find the inverse of matrix if possible: $\begin{bmatrix} 0 & 3 & 4 \\ -4 & 2 & 6 \\ 2 & 0 & 5 \end{bmatrix}$	04	OR	Convert $\cos \frac{2\theta + 2\phi}{2} \sin \frac{2\theta - 2\phi}{2}$ into sum or difference form:	04

SECTION – C (Marks 32)

Note: Solve the following Questions.

(4 x 8 = 32)

(Use of the graph paper is not necessary. Candidates can make their own grid on answer sheet)					
Q.3	a. Recognize the sequence (as A.P or G.P) and find the general term of the sequence $4, 2, 1, \frac{1}{2}, \dots$ b. Find the sum up to n terms. c. Find the sum up to infinity.	4+2 +2	OR	A committee of 5 members is to be selected from a group of 8 men and 6 women. In how many ways can the committee be formed if it must include at least 2 women?	08
Q.4	Solve: $2x + y - z = 1$ $x + 2y + z = 4$ $3x + y + 2z = 7$	08	OR	In a robotics project, a mechanical arm moves in such a way that its vertical position h (in meters) is modeled by $h(t) = 15 \cos\left(\frac{t}{2}\right)$, where t is time in seconds. Determine the first time t when the vertical position of the arm reaches 7.5 meters. Also determine the value of t for maximum height.	08
Q.5	Find the middle term(s) in the expansion of $\left(3x^2 - \frac{1}{2x}\right)^{10}$. Also find the 9 th term.	08	OR	If $\sin \alpha = \frac{3}{5}$, $\sin \beta = \frac{1}{\sqrt{5}}$, where terminal ray of α and β are in 2 nd quadrant, then find the value of $\cos(\alpha - \beta)$ and $\sin(\alpha + \beta)$.	08
Q.6	a. For what value of k the remainder of $2x^3 - kx^2 + 16x + 12$ will be zero if the polynomial is divided by $x - 2$ b. Factorize the polynomial completely.	3+5	OR	a. Prove that scalar triple product is invariant under cyclic permutation. b. Verify using vectors $\vec{i}, \vec{j}, \vec{k}$	5+3

SECTION – B (Marks 48)

Note: Candidates are to attempt 12 parts. I have provided solutions for all original and "OR" options.

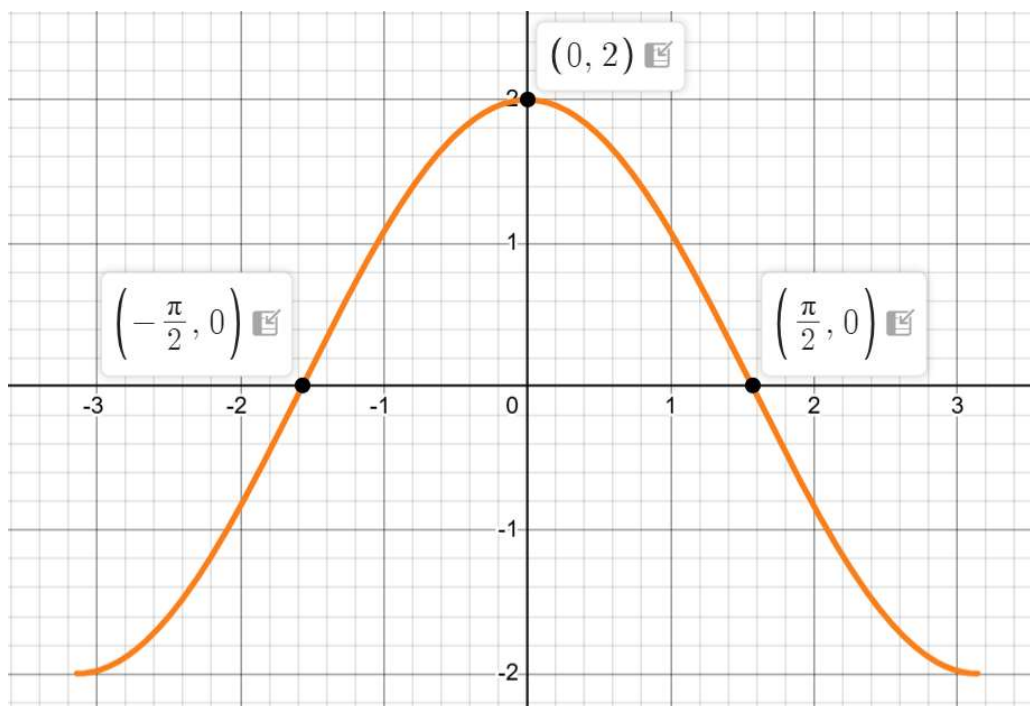
Q.2 (i)

Original: Find the sum of the geometric series $\frac{2}{3}, -1, \frac{3}{2}, \dots$ up to 5 terms.

- $a = \frac{2}{3}, r = \frac{-1}{2/3} = -\frac{3}{2}$
- Using $S_n = \frac{a(1-r^n)}{1-r}$:
- $S_5 = \frac{\frac{2}{3}(1-(-\frac{3}{2})^5)}{1-(-\frac{3}{2})} = \frac{\frac{2}{3}(1+\frac{243}{32})}{\frac{5}{2}} = \frac{2}{3} \cdot \frac{275}{32} \cdot \frac{2}{5} = \frac{55}{24}$

OR: Sketch the graph of $y = 2 \cos(x)$ for $-\pi \leq x \leq \pi$.

x	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$y = 2\cos(x)$	-2	$-\sqrt{2}$	0	$\frac{2}{\sqrt{2}}$	2	$\frac{2}{\sqrt{2}}$	0	$-\sqrt{2}$	-2



Q.2 (ii)

Original: Find the last two digits of $(12)^{11}$.

For last two digits, the number must be divisible by 100

$$(12)^{11} = (10+2)^{11}$$

$$= \binom{11}{0} 10^{11} + \binom{11}{1} 10^{10} \cdot 2^1 + \dots + \binom{11}{10} 10^1 \cdot 2^{10} + \binom{11}{11} 2^{11}$$

Here, all terms are multiples of $100 (10^2)$ except last two terms, so,

$$12^{11} = 10^2 K + \binom{11}{10} 10 \cdot 2^{10} + \binom{11}{11} 2^{11}, K \in \mathbb{Z}$$

Ignore $10^2 K$, as it is divisible by 100

$$\text{Last two digits of } 12^{11} = \binom{11}{10} 10 \cdot 2^{10} + \binom{11}{11} 2^{11}$$

$$= 11 \cdot 10 \cdot (1024) + 1 \cdot 2^{11}$$

$$= \underline{112640} + \underline{2048}$$

Hence, last two digits of $12^{11} = 40 + 48 = 88$

Ans = 88

OR: Find the rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$.

$$R \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \quad \begin{array}{l} \text{By} \\ R_2 + (-2)R_1 \rightarrow R_2' \\ R_3 + (-1)R_1 \rightarrow R_3' \end{array}$$

$$R \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{By} \\ R_2 \leftrightarrow R_3 \end{array}$$

$$R \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{By} \\ (-1)R_2 \rightarrow R_2' \end{array}$$

Thus, No. of non-zero rows = 2

Hence Rank of the matrix = 2

Q.2 (iii)

Original: Given $z_1 = 2(\cos 30^\circ + i \sin 30^\circ)$ and $z_2 = 3(\cos 60^\circ + i \sin 60^\circ)$, find $z_1 z_2$ in polar form.

For complex numbers in polar form:

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1), \quad z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$$

their product is:

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

Given:

$$z_1 = 2(\cos 30^\circ + i \sin 30^\circ)$$

$$z_2 = 3(\cos 60^\circ + i \sin 60^\circ)$$

Multiply moduli:

$$2 \times 3 = 6$$

Add arguments:

$$30^\circ + 60^\circ = 90^\circ$$

Therefore,

$$z_1 z_2 = 6(\cos 90^\circ + i \sin 90^\circ)$$

Hence, the polar form is:

$$\boxed{6(\cos 90^\circ + i \sin 90^\circ)}$$

OR: Find the value of 'n' if ${}^n P_4 : {}^{n-1} P_3 = 9 : 1$.

Given:

$${}^n P_4 : {}^{n-1} P_3 = 9 : 1$$

Using the permutation formula:

$${}^n P_r = \frac{n!}{(n-r)!}$$

$${}^nP_4 = \frac{n!}{(n-4)!} = n(n-1)(n-2)(n-3)$$

and

$${}^{n-1}P_3 = \frac{(n-1)!}{(n-4)!} = (n-1)(n-2)(n-3)$$

So,

$$\frac{{}^nP_4}{{}^{n-1}P_3} = \frac{n(n-1)(n-2)(n-3)}{(n-1)(n-2)(n-3)} = n$$

Given that the ratio equals 9,

$$n = 9$$

Therefore,

$$\boxed{9}$$

(iv) Find the remainder when 9^{205} is divided by 31 using Binomial expansion.

$$(9)^{205} = (3^2)^{205} = 3^{410}$$

We modify 3 in the form of 31, so
 Multiples of 31 : 31, 62, 93, 124, 155, 186, 217, **248**

Powers of 3 : 3, 9, 81, **243**

248 and 243 are close ones, so,

$$243 = 3^5, \quad 248 = 31 \times 8$$

Thus,

$$9^{205} = (3^2)^{410} = (3^5)^{82} = (243)^{82} = (248 - 5)^{82}$$

∴ Binomial theorem,

By using Binomial theorem,

$$(248 + (-5))^{82} = \binom{82}{0} 248^0 + \binom{82}{1} 248^1 (-5)^1 + \dots + \binom{82}{81} 248^1 (-5)^1 + \binom{82}{82} (-5)^{82}$$

Note that all terms are the multiples of 31 except the last one as 248 is divisible by 31

$$\Rightarrow 9^{205} = 31k + \binom{82}{82} (-5)^{82}, k \in \mathbb{Z}$$

Ignore $31k$ as it is divisible by 31

Remainder of 9^{205} depends on $(-5)^{82}$

$$(-5)^{82} = (5)^{82}$$

$$\text{As } 5^1 = 5, 5^2 = 25$$

The cycle length of 5 is 1 only

... remainder of 9^{205} is 5

Hence $\boxed{\text{Remainder of } 9^{205} = 5} \text{ Ans}$

OR Find all roots of: $x^3 - 3x^2 - 4x + 12$ by Factor theorem.

$$P = 12 : \pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$$

$$Q = 1 : \pm 1$$

The possible roots are

$$\frac{P}{Q} : \pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$$

On checking, $x = 2$ satisfies the given polynomial as $P(2) = 0$.

Thus $(x - 2)$ is a factor of $P(x)$

Now

$$\begin{array}{r} x^2 - x - 6 \\ x-2 \overline{) x^3 - 3x^2 - 4x + 12} \\ \underline{\ominus x^3 - 2x^2} \\ -x^2 - 4x + 12 \\ \underline{\ominus x^2 + 2x} \\ -6x + 12 \\ \underline{+6x - 12} \\ 0 \end{array}$$

Note: You can

Note: You can do it by synthetic division as well.

$$\begin{array}{r} \overline{+x^2 + 2x} \\ -6x + 12 \\ \hline +6x + 12 \\ \hline 0 \end{array}$$

As $Q(x) = x^2 - x - 6$ so

$Q(x) = (x-3)(x+2)$ are the missing factors

Hence $\boxed{x^3 - 3x^2 - 4x + 12 = (x-2)(x+2)(x-3)}$

OR the roots of $x^3 - 3x^2 - 4x + 12$ are $\boxed{2, -2, 3}$ Ans

The text from the image is as follows:

(v) If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ then find A_{13} , A_{23} , A_{31} and A_{33} .

To find the cofactors A_{ij} , we use the formula $A_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the determinant of the minor matrix.

1. Find A_{13} :

$$M_{13} = \begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix} = (12 - 12) = 0$$

$$A_{13} = (-1)^{1+3}(0) = 0$$

2. Find A_{23} :

$$M_{23} = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = (6 - 6) = 0$$

$$A_{23} = (-1)^{2+3}(0) = 0$$

3. Find A_{31} :

$$M_{31} = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = (12 - 12) = 0$$

$$A_{31} = (-1)^{3+1}(0) = 0$$

4. Find A_{33} :

$$M_{33} = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = (4 - 4) = 0$$

$$A_{33} = (-1)^{3+3}(0) = 0$$

The values for A_{13} , A_{23} , A_{31} , and A_{33} are all 0.

OR Prove by mathematical induction that: $2^n \geq n + 1$ for all $n \geq 1$

Step 1: Basis Step

For $n = 1$:

$$2^1 \geq 1 + 1$$

$$2 \geq 2$$

The statement is **true** for $n = 1$.

Step 2: Inductive Step

Assume the statement is true for $n = k$, where k is some positive integer:

$$2^k \geq k + 1 \quad \dots \text{(Equation 1)}$$

We need to prove the statement is true for $n = k + 1$, i.e., $2^{k+1} \geq (k + 1) + 1$, which simplifies to $2^{k+1} \geq k + 2$.

Starting with the left side of the inequality for $n = k + 1$:

$$2^{k+1} = 2 \cdot 2^k$$

From our hypothesis (Equation 1), we know $2^k \geq k + 1$. Multiplying both sides by 2:

$$2 \cdot 2^k \geq 2(k + 1)$$

$$2^{k+1} \geq 2k + 2$$

Now, observe that $2k + 2$ can be rewritten as $k + (k + 2)$. Since $k \geq 1$, it is clear that:

$$2k + 2 > k + 2$$

Therefore:

$$2^{k+1} \geq k + 2$$

Since the basis step and the inductive step are both proven, by the principle of mathematical induction, $2^n \geq n + 1$ is true for all $n \geq 1$.

(vi) Solve the following equations:

$$(2 + i)\omega = 1$$

$$(3 - i)z + 5\omega = 2 - i$$

Note:- This q/s. can be solved by Cramer's Rule, matrix inversion method as well from Elimination method.
** I am solving it using substitution method.*

Step 1: Find ω

$$\omega = \frac{1}{2+i}$$

Rationalize:

$$\omega = \frac{1}{2+i} \cdot \frac{2-i}{2-i} = \frac{2-i}{(2+i)(2-i)}$$

Denominator:

$$(2+i)(2-i) = 4 + 1 = 5$$

So:

$$\omega = \frac{2-i}{5}$$

Step 2: Substitute into second equation

$$(3-i)z + 5\omega = 2-i$$

Substitute ω :

$$5\omega = 5 \cdot \frac{2-i}{5} = 2-i$$

So the equation becomes:

$$(3-i)z + (2-i) = 2-i$$

Step 3: Solve for z

Subtract $2-i$ from both sides:

$$(3-i)z = 0$$

Since $3-i \neq 0$, we get:

$$z = 0$$

Final Answers:

$$\boxed{\omega = \frac{2-i}{5}, \quad z = 0}$$



OR A force $F = 3i - 2j + k$ acts at point $(1, 2, 3)$. Find the moment about the origin.

OR A force $\vec{F} = 3\hat{i} - 2\hat{j} + \hat{k}$ acts at point $(1, 2, 3)$. Find the moment about the origin.

$\vec{M} = \vec{r} \times \vec{F}$
 $\vec{r} = \vec{OP}$
 $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k}$

Given, $\vec{F} = 3\hat{i} - 2\hat{j} + \hat{k}$

$$\vec{M} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & -2 & 1 \end{vmatrix}$$

$$= \hat{i}(2+6) - \hat{j}(1-9) + \hat{k}(-2-6)$$

$$\vec{M} = 8\hat{i} + 8\hat{j} - 8\hat{k}$$

Ans

(vii) Verify the identity: $\cos(3\alpha) = 4\cos^3(\alpha) - 3\cos(\alpha)$

(b) $\cos(3\alpha) = \cos(2\alpha + \alpha) = \cos(2\alpha)\cos\alpha - \sin(2\alpha)\sin\alpha$

$$= (2\cos^2\alpha - 1)\cos\alpha - (2\sin\alpha\cos\alpha)\sin\alpha \quad \dots \text{by (4) and (6)}$$

$$= 2\cos^3\alpha - \cos\alpha - 2\sin^2\alpha\cos\alpha$$

$$= 2\cos^3\alpha - \cos\alpha - 2(1 - \cos^2\alpha)\cos\alpha \quad \dots (\because \sin^2\alpha + \cos^2\alpha = 1)$$

$$= 2\cos^3\alpha - \cos\alpha - 2\cos\alpha + 2\cos^3\alpha$$

$$\cos(3\alpha) = 4\cos^3\alpha - 3\cos\alpha \quad (19)$$

OR A class has 5 boys and 4 girls. In how many ways can a group of 3 students be formed?

We have:

- 5 boys + 4 girls = 9 students total
- We need to form a group of 3 students
- No restrictions given, so order does not matter → use combinations.

Step 1: Use combination formula

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Step 2: Apply values

$$\begin{aligned} {}^9C_3 &= \frac{9!}{3!6!} \\ &= \frac{9 \cdot 8 \cdot 7}{3 \cdot 2 \cdot 1} \end{aligned}$$

Step 3: Simplify

$$= \frac{504}{6} = 84$$

(viii) Find the volume of the parallelepiped: $a = i + j + 2k, b = 2i + j - k, c = i + k$

Volume of parallelepiped = $|\vec{a} \cdot (\vec{b} \times \vec{c})|$

$$|\vec{a} \cdot \vec{b} \times \vec{c}| = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 1 & -1 \\ 1 & 0 & 1 \end{vmatrix}$$

$$= 1(1-0) - 1(2+1) + 2(0-1)$$

$$= 1 - 3 - 2 = -4$$

Since $|\vec{a} \cdot \vec{b} \times \vec{c}|$ is the magnitude,

$\Rightarrow$ Volume of parallelepiped = 4 cubic units

$\rightarrow$ You can also do it by $f(x) \div (x+2)$ Any

(ix) Determine whether $(x + 2)$ is a factor of: $f(x) = x^3 + 2x^2 - 4x - 8$

To check whether $(x + 2)$ is a factor of

$$f(x) = x^3 + 2x^2 - 4x - 8$$

we use the Factor Theorem:

Substitute $x = -2$

$$f(-2) = (-2)^3 + 2(-2)^2 - 4(-2) - 8$$

$$= -8 + 2(4) + 8 - 8$$

$$= -8 + 8 + 8 - 8$$

$$= 0$$

Since:

$$f(-2) = 0$$

$(x + 2)$ is a factor of $f(x)$

(x)

(x)

OR Find the value of " λ " so that vectors $v = \hat{i} - \hat{j} + \hat{k}$, $u = 2\hat{i} + \hat{j} + \hat{k}$ and $w = -\hat{i} + \lambda\hat{j} + 2\hat{k}$ are coplanar.

Three vectors are coplanar if $[\vec{v} \cdot (\vec{u} \times \vec{w})] = 0$. This is calculated using the determinant of their components:

$$\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & 1 \\ -1 & \lambda & 2 \end{vmatrix} = 0 \Rightarrow 1(2-\lambda) + 1(4+1) + 1(2\lambda+1) = 0$$
$$\Rightarrow 2-\lambda + 5 + 2\lambda + 1 = 0$$
$$\Rightarrow \lambda + 8 = 0$$
$$\Rightarrow \boxed{\lambda = -8} \text{ Ans}$$

(viii)

OR Find the maximum and minimum values of $y = \frac{1}{4+5 \cos 3x}$

$$\text{As } -1 \leq \cos 3x \leq 1$$

$$\Rightarrow -5 \leq 5 \cos 3x \leq 5$$

$$\text{Minimum value of } y: m = \frac{1}{4-5}$$

$$\Rightarrow \boxed{m = -1}$$

$$\text{Maximum value of } y: M = \frac{1}{4+5}$$

Maximum value of y is $4+5$

$$\Rightarrow \boxed{M = \frac{1}{9}}$$

Thus, $\boxed{\begin{array}{l} \text{max. value of } y = \frac{1}{9} \\ \text{min value of } y = -1 \end{array}}$ *Ans*

(ix)

OR Find the sum to n terms of the sequence: 3, 9, 27, 81, ...

The given sequence is:

$$3, 9, 27, 81, \dots$$

This is a geometric progression (G.P.) with:

- First term: $a = 3$
- Common ratio: $r = \frac{9}{3} = 3$

Step 1: Use sum of first n terms of G.P.

$$S_n = a \frac{r^n - 1}{r - 1}$$

Step 2: Substitute values

$$S_n = 3 \cdot \frac{3^n - 1}{3 - 1}$$

$$S_n = 3 \cdot \frac{3^n - 1}{2}$$

Final Answer:

$$\boxed{S_n = \frac{3(3^n - 1)}{2}}$$

(x) Determine the properties of the graph of: $y = \cos x$ i.e. Domain, Range, Period and Amplitude.

- i) Domain of $y = D_y = \mathbb{R}$ or $(-\infty, +\infty)$
- ii) Range of $y = R_y = [-1, +1]$
- iii) Period = $T = \frac{2\pi}{|1|} = 2\pi$
- iv) Amplitude = $A = |1| = 1$

(xi) Find the Sum of first 20 terms of A.P with $a = 1, d = 3$.

We are given an A.P.:

- First term $a = 1$
- Common difference $d = 3$
- Number of terms $n = 20$

Step 1: Use sum formula of A.P.

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

Step 2: Substitute values

$$\begin{aligned} S_{20} &= \frac{20}{2} [2(1) + (20 - 1)(3)] \\ &= 10 [2 + 19 \cdot 3] \\ &= 10 [2 + 57] \\ &= 10 \cdot 59 \end{aligned}$$

Final Answer:

590

OR Verify: $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

Example 1: Prove that:

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Solution: $\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)}$

$$\frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}$$

Dividing numerator and denominator by $\cos \alpha \cos \beta$, we get:

$$\frac{\frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta}}{\frac{\cos \alpha \cos \beta - \sin \alpha \sin \beta}{\cos \alpha \cos \beta}} = \frac{\frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} + \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta}}{\frac{\cos \alpha \cos \beta}{\cos \alpha \cos \beta} - \frac{\sin \alpha \sin \beta}{\cos \alpha \cos \beta}}$$

$$= \frac{\frac{\sin \alpha}{\cos \alpha} + \frac{\sin \beta}{\cos \beta}}{1 - \frac{\sin \alpha}{\cos \alpha} \times \frac{\sin \beta}{\cos \beta}} = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\therefore \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Prove that:

(i) $\sin(-\alpha) = -\sin \alpha$

(ii) $\tan(-\alpha) = -\tan \alpha$

(iii) $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$

Key Facts

The identities of tangent ratio are true for all values of α and β for which $\cos \alpha \neq 0$ and $\cos \beta \neq 0$, and for which $\tan(\alpha + \beta)$ or $\tan(\alpha - \beta)$ are defined.

(xii) Find the inverse of matrix if possible:

$$A = \begin{bmatrix} 0 & 3 & 4 \\ -4 & 2 & 6 \\ 2 & 0 & 5 \end{bmatrix}$$

Step 1: Find determinant of A

$$|A| = \begin{vmatrix} 0 & 3 & 4 \\ -4 & 2 & 6 \\ 2 & 0 & 5 \end{vmatrix}$$

$$\begin{aligned}
&= -3[(-4)(5) - 6(2)] + 4[(-4)(0) - 2(2)] \\
&= -3[-20 - 12] + 4[0 - 4] \\
&= -3(-32) + 4(-4) \\
&= 96 - 16 \\
&= 80
\end{aligned}$$

Since:

$$|A| \neq 0$$

the inverse exists.

2. Find the Cofactors (A_{ij})

We calculate $A_{ij} = (-1)^{i+j} M_{ij}$ for each element:

- $A_{11}: + \begin{vmatrix} 2 & 6 \\ 0 & 5 \end{vmatrix} = 10 - 0 = 10$
- $A_{12}: - \begin{vmatrix} -4 & 6 \\ 2 & 5 \end{vmatrix} = -(-20 - 12) = 32$
- $A_{13}: + \begin{vmatrix} -4 & 2 \\ 2 & 0 \end{vmatrix} = 0 - 4 = -4$
- $A_{21}: - \begin{vmatrix} 3 & 4 \\ 0 & 5 \end{vmatrix} = -(15 - 0) = -15$
- $A_{22}: + \begin{vmatrix} 0 & 4 \\ 2 & 5 \end{vmatrix} = 0 - 8 = -8$
- $A_{23}: - \begin{vmatrix} 0 & 3 \\ 2 & 0 \end{vmatrix} = -(0 - 6) = 6$
- $A_{31}: + \begin{vmatrix} 3 & 4 \\ 2 & 6 \end{vmatrix} = 18 - 8 = 10$
- $A_{32}: - \begin{vmatrix} 0 & 4 \\ -4 & 6 \end{vmatrix} = -(0 - (-16)) = -16$
- $A_{33}: + \begin{vmatrix} 0 & 3 \\ -4 & 2 \end{vmatrix} = 0 - (-12) = 12$

3. Find the Adjoint $\text{adj}(A)$

The adjoint is the transpose of the cofactor matrix:

$$\text{adj}(A) = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} = \begin{bmatrix} 10 & -15 & 10 \\ 32 & -8 & -16 \\ -4 & 6 & 12 \end{bmatrix}$$

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Step 4: Use inverse formula

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

$$A^{-1} = \frac{1}{80} \begin{bmatrix} 10 & -15 & 10 \\ 32 & -8 & -16 \\ -4 & 6 & 12 \end{bmatrix}$$

Simplifying:

$$A^{-1} = \begin{bmatrix} \frac{1}{8} & -\frac{3}{16} & \frac{1}{8} \\ \frac{2}{5} & -\frac{1}{10} & -\frac{1}{5} \\ -\frac{1}{20} & \frac{3}{40} & \frac{3}{20} \end{bmatrix}$$

OR Convert $\cos \frac{2\theta+2\phi}{2} \sin \frac{2\theta-2\phi}{2}$ into sum or difference form.

$$= \cos(\theta + \phi) \sin(\theta - \phi)$$

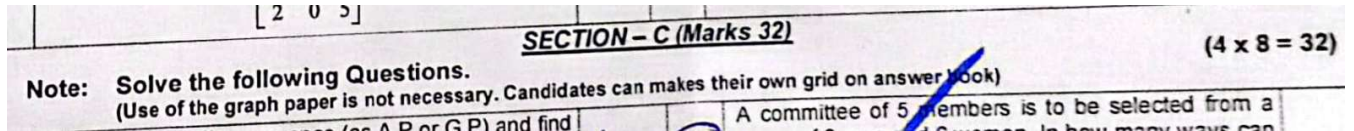
$$= \frac{1}{2} 2 \cos(\theta + \phi) \sin(\theta - \phi)$$

$$= \frac{1}{2} [\sin(\theta + \phi + \theta - \phi) - \sin(\theta + \phi - \theta + \phi)]$$

$$= \frac{1}{2} [\sin 2\theta - \sin 2\phi]$$

$$\boxed{\sin 2\theta - \sin 2\phi} \text{ Ans}$$

$$= \boxed{\frac{\sin 2\alpha - \sin 2\alpha}{2}} \quad \text{Ans}$$



Q.3 a. Recognize the sequence (as A.P or G.P) and find the general term of the sequence

$$4, 2, 1, \frac{1}{2}, \dots$$

b. Find the sum up to n terms.

c. Find the sum up to infinity.

- **Check Ratio:** $\frac{2}{4} = \frac{1}{2}$ and $\frac{1}{2}$. Since the ratio is constant, it is a **Geometric Progression (G.P.)**.
- **First term (a):** 4
- **Common ratio (r):** $\frac{1}{2}$
- **General Term (a_n):** $a_n = ar^{n-1} \Rightarrow a_n = 4\left(\frac{1}{2}\right)^{n-1} = 2^2 \cdot 2^{-(n-1)} = 2^{3-n}$.

b. Sum up to n terms (S_n):

Since $r < 1$:

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{4(1 - (\frac{1}{2})^n)}{1 - \frac{1}{2}} = \frac{4(1 - \frac{1}{2^n})}{\frac{1}{2}} = 8(1 - \frac{1}{2^n})$$

c. Sum to infinity (S_∞):

$$S_\infty = \frac{a}{1 - r} = \frac{4}{1 - \frac{1}{2}} = \frac{4}{\frac{1}{2}} = 8$$

OR A committee of 5 members is to be selected from a group of 8 men and 6 women. In how many ways can the committee be formed if it must include at least 2 women?

We have:

- 8 men
- 6 women
- Committee of 5 members
- At least 2 women

We count all possible valid cases.

Case 1: Exactly 2 women

Choose 2 women from 6 and 3 men from 8:

$$\begin{aligned} & {}^6C_2 \cdot {}^8C_3 \\ &= 15 \times 56 = 840 \end{aligned}$$

Case 2: Exactly 3 women

Choose 3 women from 6 and 2 men from 8:

$$\begin{aligned} & {}^6C_3 \cdot {}^8C_2 \\ &= 20 \times 28 = 560 \end{aligned}$$

Case 3: Exactly 4 women

Choose 4 women from 6 and 1 man from 8:

$$\begin{aligned} & {}^6C_4 \cdot {}^8C_1 \\ &= 15 \times 8 = 120 \end{aligned}$$

Case 4: Exactly 5 women

Choose 5 women from 6:

$$\begin{aligned} & {}^6C_5 \cdot {}^8C_0 \\ &= 6 \times 1 = 6 \end{aligned}$$

Total ways

$$\begin{aligned}840 + 560 + 120 + 6 \\ = 1526\end{aligned}$$

Final Answer

1526

Q.4 Solve:

$$2x + y - z = 1$$

$$x + 2y + z = 4$$

$$3x + y + 2z = 7$$

 Must verify by
the calculator
at the end.

Step 1: Find the Determinant of the Coefficient Matrix (D)

$$D = \begin{vmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ 3 & 1 & 2 \end{vmatrix}$$

Expanding along the first row:

$$D = 2 \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 3 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$$

$$D = 2(4 - 1) - 1(2 - 3) - 1(1 - 6)$$

$$D = 2(3) - 1(-1) - 1(-5)$$

$$D = 6 + 1 + 5 = 12$$

Step 2: Find D_x , D_y , and D_z

Replace the respective column with the constants $[1, 4, 7]$:

For D_x :

$$D_x = \begin{vmatrix} 1 & 1 & -1 \\ 4 & 2 & 1 \\ 7 & 1 & 2 \end{vmatrix} = 1(4 - 1) - 1(8 - 7) - 1(4 - 14) = 3 - 1 + 10 = 12$$

For D_y :

$$D_y = \begin{vmatrix} 2 & 1 & -1 \\ 1 & 4 & 1 \\ 3 & 7 & 2 \end{vmatrix} = 2(8 - 7) - 1(2 - 3) - 1(7 - 12) = 2(1) + 1 + 5 = 8$$

For D_z :

$$D_z = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 4 \\ 3 & 1 & 7 \end{vmatrix} = 2(14 - 4) - 1(7 - 12) + 1(1 - 6) = 2(10) + 5 - 5 = 20$$

Step 3: Calculate the Variables

Using the formulas $x = \frac{D_x}{D}$, $y = \frac{D_y}{D}$, and $z = \frac{D_z}{D}$:

- $x = \frac{12}{12} = 1$
- $y = \frac{8}{12} = \frac{2}{3}$
- $z = \frac{20}{12} = \frac{5}{3}$

Final Answer

The solution to the system is:

$$x = 1, \quad y = \frac{2}{3}, \quad z = \frac{5}{3}$$

OR In a robotics project, a mechanical arm moves in such a way that its vertical position h (in meters) is modeled by $h(t) = 15 \cos\left(\frac{t}{2}\right)$, where t is time in seconds. Determine the first time t when the vertical position of the arm reaches 7.5 meters. Also determine the value of t for maximum height.

We are given:

$$h(t) = 15 \cos \left(\frac{t}{2} \right)$$

1) First time when height is 7.5 m

Set:

$$15 \cos \left(\frac{t}{2} \right) = 7.5$$

Divide by 15:

$$\cos \left(\frac{t}{2} \right) = \frac{1}{2}$$

We know:

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3} \text{ (first solution)}$$

So:

$$\frac{t}{2} = \frac{\pi}{3}$$

Multiply by 2:

$$t = \frac{2\pi}{3}$$

Using the value of $\pi \approx 3.14159$:

$t \approx 2.09$ seconds.

2) Time when height is maximum

Maximum value of cosine is:

$$\cos x = 1$$

So:

$$h_{\max} = 15$$

This occurs when:

$$\frac{t}{2} = 0, 2\pi, 4\pi, \dots$$

First time:

$$\frac{t}{2} = 0 \Rightarrow t = 0$$

Maximum height time:

$t = 0 \text{ seconds (first occurrence)}$

Time for maximum height: 0 seconds (initial position) and every 4π seconds thereafter.

Q.5 Find the middle term(s) in the expansion of $(3x^2 - \frac{1}{2x})^{10}$. Also find the 9th term.

Solution

For a binomial expansion of $(a + b)^n$:

- If n is even, there is one middle term at position $\frac{n}{2} + 1$.
- The general term is given by: $T_{r+1} = \binom{n}{r} a^{n-r} b^r$.

In this problem: $a = 3x^2$, $b = -\frac{1}{2x}$, and $n = 10$.

1. Find the Middle Term

Since $n = 10$ is even, the middle term is the $(\frac{10}{2} + 1) = 6^{\text{th}}$ term (T_6).

For T_6 , we use $r = 5$:

$$T_6 = \binom{10}{5} (3x^2)^{10-5} \left(-\frac{1}{2x}\right)^5$$

$$T_6 = 252 \cdot (3x^2)^5 \cdot \left(-\frac{1}{32x^5}\right)$$

$$T_6 = -\frac{61236}{32}x^5 = -\frac{15309}{8}x^5$$

2. Find the 9th Term (T_9)

For the 9th term, we use $r = 8$:

$$T_9 = \binom{10}{8} (3x^2)^{10-8} \left(-\frac{1}{2x}\right)^8$$

$$T_9 = 45 \cdot (3x^2)^2 \cdot \left(\frac{1}{256x^8}\right)$$

$$T_9 = 45 \cdot 9x^4 \cdot \frac{1}{256x^8}$$

$$T_9 = \frac{405}{256x^4}$$

Final Answers:

- **Middle Term (T_6):** $-\frac{15309}{8}x^5$
- **9th Term (T_9):** $\frac{405}{256x^4}$

OR If $\sin \alpha = \frac{3}{5}$, $\sin \beta = \frac{1}{\sqrt{5}}$, where terminal ray of α and β are in 2nd quadrant, then find the value of $\cos(\alpha - \beta)$ and $\sin(\alpha + \beta)$.

To find the required values, we first need to determine $\cos \alpha$ and $\cos \beta$. Since both angles are in the 2nd **quadrant**, the cosine values must be **negative**.

1. Find $\cos \alpha$ and $\cos \beta$

Using the identity $\cos^2 \theta + \sin^2 \theta = 1$:

- **For α :**

$$\cos \alpha = -\sqrt{1 - \sin^2 \alpha} = -\sqrt{1 - \left(\frac{3}{5}\right)^2} = -\sqrt{1 - \frac{9}{25}} = -\sqrt{\frac{16}{25}} = -\frac{4}{5}$$

- **For β :**

$$\cos \beta = -\sqrt{1 - \sin^2 \beta} = -\sqrt{1 - \left(\frac{1}{\sqrt{5}}\right)^2} = -\sqrt{1 - \frac{1}{5}} = -\sqrt{\frac{4}{5}} = -\frac{2}{\sqrt{5}}$$

2. Find $\cos(\alpha - \beta)$

Using the formula: $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

$$\cos(\alpha - \beta) = \left(-\frac{4}{5}\right) \left(-\frac{2}{\sqrt{5}}\right) + \left(\frac{3}{5}\right) \left(\frac{1}{\sqrt{5}}\right)$$

$$\cos(\alpha - \beta) = \frac{8}{5\sqrt{5}} + \frac{3}{5\sqrt{5}} = \frac{11}{5\sqrt{5}}$$

3. Find $\sin(\alpha + \beta)$

Using the formula: $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

$$\sin(\alpha + \beta) = \left(\frac{3}{5}\right) \left(-\frac{2}{\sqrt{5}}\right) + \left(-\frac{4}{5}\right) \left(\frac{1}{\sqrt{5}}\right)$$

$$\sin(\alpha + \beta) = -\frac{6}{5\sqrt{5}} - \frac{4}{5\sqrt{5}} = -\frac{10}{5\sqrt{5}} = -\frac{2}{\sqrt{5}}$$

Final Answers:

- $\cos(\alpha - \beta) = \frac{11}{5\sqrt{5}}$
- $\sin(\alpha + \beta) = -\frac{2}{\sqrt{5}}$

Q.6 a. For what value of k the remainder of $2x^3 - kx^2 + 16x + 12$ will be zero if the polynomial is divided by $x - 2$

b. Factorize the polynomial completely.

Part (a): Find the value of k

According to the **Remainder Theorem**, if a polynomial $P(x)$ is divided by $(x - a)$, the remainder is $P(a)$. Here, for the remainder to be zero, $P(2)$ must equal 0.

$$\text{Let } P(x) = 2x^3 - kx^2 + 16x + 12.$$

$$\text{Set } P(2) = 0:$$

$$2(2)^3 - k(2)^2 + 16(2) + 12 = 0$$

$$2(8) - k(4) + 32 + 12 = 0$$

$$16 - 4k + 44 = 0$$

$$60 - 4k = 0$$

$$4k = 60$$

$$k = 15$$

The value of k is 15.

Part (b): Factorize the polynomial completely

Now that we know $k = 15$, the polynomial is $P(x) = 2x^3 - 15x^2 + 16x + 12$. Since the remainder was zero when divided by $(x - 2)$, we know $(x - 2)$ is a factor.

By Synthetic division,

$$\begin{array}{r|rrrr} 2 & 2 & -15 & 16 & 12 \\ & \downarrow & 4 & -22 & -12 \\ \hline & 2 & -11 & -6 & 0 \end{array}$$

$$\begin{array}{r|rrrr} 2 & -11 & -6 & 0 \end{array}$$

Thus, $Q(x) = 2x^2 - 11x - 6$

2. Factor the Quadratic Quotient:

We need to factor $2x^2 - 11x - 6$ by splitting the middle term (product = -12 , sum = -11):

$$2x^2 - 12x + x - 6$$

$$2x(x - 6) + 1(x - 6)$$

$$(2x + 1)(x - 6)$$

Final Factorized Form:

$$P(x) = (x - 2)(2x + 1)(x - 6)$$

OR

a. Prove that scalar triple product is invariant under cyclic permutation.

b. Verify using vectors i, j, k

Part (a): Proof of Cyclic Invariance

The scalar triple product of three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is defined as $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. In component form, if:

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$$

$$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$$

$$\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$$

Then the scalar triple product is given by the determinant:

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Step 1: Perform the first cyclic shift

According to the properties of determinants, swapping two rows changes the sign. To shift **a, b, c** to **b, c, a**, we perform two swaps:

$$1. \text{ Swap Row 1 and Row 2: } - \begin{vmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$2. \text{ Swap Row 2 and Row 3: } (-1)(-1) \begin{vmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = [\mathbf{b}, \mathbf{c}, \mathbf{a}]$$

Step 2: Perform the second cyclic shift

Similarly, swapping rows again to move **b** to the bottom:

$$1. \text{ Swap Row 1 and Row 2: } - \begin{vmatrix} c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$2. \text{ Swap Row 2 and Row 3: } (-1)(-1) \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = [\mathbf{c}, \mathbf{a}, \mathbf{b}]$$

Conclusion:

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}] = [\mathbf{c}, \mathbf{a}, \mathbf{b}]$$

This proves that the scalar triple product is invariant under cyclic permutation.

Part (b): Verification using i, j, k

Let **a = i**, **b = j**, and **c = k**.

1. Calculate $[\mathbf{i}, \mathbf{j}, \mathbf{k}]$:

$$\mathbf{i} \cdot (\mathbf{j} \times \mathbf{k}) = \mathbf{i} \cdot \mathbf{i} = 1$$

2. Calculate $[\mathbf{j}, \mathbf{k}, \mathbf{i}]$:

$$\mathbf{j} \cdot (\mathbf{k} \times \mathbf{i}) = \mathbf{j} \cdot \mathbf{j} = 1$$

3. Calculate $[\mathbf{k}, \mathbf{i}, \mathbf{j}]$:

$$\mathbf{k} \cdot (\mathbf{i} \times \mathbf{j}) = \mathbf{k} \cdot \mathbf{k} = 1$$

Since $1 = 1 = 1$, the property is verified for the standard unit vectors.

	Questions	A	B	C	D
1	Period of $\tan \frac{5x}{7}$ is:	$\frac{5\pi}{7}$	5π	$\frac{7\pi}{5}$	7π
2	A function is periodic if:	$f(x)=f(-x)$	$f(x)=1$	$f(x)=0$	$f(x+T)=f(x)$
3	Range of $a + b\sin 2\theta$ is:	$a \pm b$	$a + b$	$a - b$	$2(a \pm b)$
4	Number of ways of arrangements of the letters in the word "SUNDAY" are:	120	480	600	720
5	Number of ways to select 8 players from 12 are:	495	720	11885	19958
6	Factorize $z^2 + 289$:	$(z + 17)$ $(z - 17)$	$(z + 17i)$ $(z + 17i)$	$(z + 17i)$ $(z - 17i)$	$(z - 17i)$ $(z - 17i)$
7	Argument of $z = \sqrt{3} + i$ is:	$\frac{\pi}{3}$	$\frac{\pi}{4}$	$\frac{\pi}{9}$	$\frac{\pi}{6}$
8	Rank of matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{bmatrix}$ is:	1	2	3	4
9	A consistent system MUST have at least:	At least one solution	Infinite solutions	No solution	Maximum two solutions
10	Formula of n^{th} term of an A.P is:	$a + nd$	$a + (n - 1)d$	$a + (n + 1)d$	$a(n - 1)d$
11	H.P decrease when A.P is:	Decreases	Increases	Constant	Zero
12	Binomial expansion uses:	Factorization	Roots	Division	Pascal triangle
13	In mathematical induction assume true for $n=k$ is:	Base case	Proof	Hypothesis	Result
14	In polynomial division Degree of remainder is:	Greater	Equal	Less than divisor	Infinite
15	If $(x-2)$ is factor of $f(x)$, then $f(2) =$	-2	0	1	2
16	Scalar product of two non-zero vector is maximum when both vectors are:	Parallel	In opposite direction	Mutually Perpendicular	At 45° to Each other
17	If vector $\vec{u} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{v} = -2\hat{i} - \hat{k}$ then the dot Product of $\vec{u}$ and $\vec{v}$ will be:	6	5	0	-2
18	Three vectors $\vec{P}, \vec{Q}$ and $\vec{R}$ are coplanar if:	$\vec{P} \times (\vec{Q} \times \vec{R}) = 0$	$(\vec{P} \cdot \vec{Q}) \times \vec{R} = 0$	$(\vec{P} \cdot \vec{Q}) \cdot \vec{R} = 0$	$\vec{P} \cdot (\vec{Q} \times \vec{R}) = 0$
19	$\cos(A - B) =$	$\cos A \cos B - \sin A \sin B$	$\cos A \cos B + \sin A \sin B$	$\cos A \sin B + \sin A \cos B$	$\cos A \sin B - \sin A \cos B$
20	Using Product to sum or difference formula $\sin 60^\circ \sin 30^\circ =$	$\frac{\sqrt{3}}{4}$	$\frac{\sqrt{3}}{2}$	1	-1



Mathematics HSSC – I (2026)

SECTION – A (20 marks) Answer key:

Scientia Vision School & College



Q.1: Fill the relevant bubble against each question according to curriculum:

S. No	Question	A	B	C	D	A	B	C	D
1.	Three vectors $\vec{P}, \vec{Q}$ and $\vec{R}$ are coplanar if:	$(\vec{P} \cdot \vec{Q}) \times \vec{R} = 0$	$(\vec{P} \cdot \vec{Q}) \cdot R = 0$	$\vec{P} \cdot (\vec{Q} \times \vec{R}) = 0$	$\vec{P} \times (\vec{Q} \times \vec{R}) = 0$	☐	☐	☒	☐
2.	$\cos(A - B) =$	$\cos A \cos B + \sin A \sin B$	$\cos A \sin B + \sin A \cos B$	$\cos A \sin B - \sin A \cos B$	$\cos A \cos B - \sin A \sin B$	☒	☐	☐	☐
3.	Using product to sum or difference formula $\sin 60^\circ \sin 30^\circ =$	$\frac{\sqrt{3}}{2}$	1	-1	$\frac{\sqrt{3}}{4}$	☐	☐	☐	☒
4.	Period of $\tan \frac{5x}{7}$ is:	5π	$\frac{7\pi}{5}$	7π	$\frac{5\pi}{7}$	☐	☒	☐	☐
5.	A function is periodic if:	$f(x) = 1$	$f(x) = 0$	$f(x+T) = f(x)$	$f(x) = f(-x)$	☐	☐	☒	☐
6.	Range of $a + b \sin 2\theta$ is:	$a + b$	$a - b$	$2(a \pm b)$	$a \pm b$	☐	☐	☐	☒
7.	Number of ways of arrangements of the letters in the word "SUNDAY" are:	480	600	720	120	☐	☐	☒	☐
8.	Number of ways to select 8 players from 12 are:	720	11885	19958	495	☐	☐	☐	☒
9.	Factorize: $z^2 + 289$:	$(z + 17i)(z + 17i)$	$(z + 17i)(z - 17i)$	$(z - 17i)(z - 17i)$	$(z + 17)(z - 17)$	☐	☒	☐	☐
10.	Argument of $z = \sqrt{3} + i$ is:	$\frac{\pi}{4}$	$\frac{\pi}{9}$	$\frac{\pi}{6}$	$\frac{\pi}{3}$	☐	☐	☒	☐

Question:	A	B	C	D	A B C D
11. Rank of matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{bmatrix}$ is:	2	3	4	1	☒ ☐ ☐ ☐
12. A consistent system MUST have at least:	Infinite solutions	No solution	Maximum two solutions	At least one solution	☐ ☐ ☐ ☒
13. Formula n^{th} term of an A.P is:	$a + (n-1)d$	$a + (n+1)d$	$a(n-1)d$	$a + nd$	☒ ☐ ☐ ☐
14. H.P decreases when A.P	Increases	Constant	Zero	Decreases	☒ ☐ ☐ ☐
15. Binomial expansion uses:	Roots	Division	✓ Pascal triangle	factorization	☐ ☒ ☐ ☐
16. In mathematical induction assume true for $n = k$ is:	Proof	Hypothesis	Result	Base case	☐ ☒ ☐ ☐
17. In polynomial division degree of remainder is:	Equal	Less than divisor	Infinite	Greater	☐ ☒ ☐ ☐
18. If $(x-2)$ is factor of $f(x)$, then $f(2) =$	0	1	2	-2	☒ ☐ ☐ ☐
19. Scalar product of two non-zero vector is maximum when both vectors are:	In opposite direction	Mutually perpendicular	At 45° to each other	Parallel	☐ ☐ ☐ ☒
20. If vector $\vec{u} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{v} = 2\hat{i} - \hat{k}$ then the dot product of $\vec{u}$ and $\vec{v}$ will be:	5	0	-2	6	☐ ☒ ☐ ☐